gpeyt1-20 · Pseudostatic infinite slope

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A long, dry, cohesionless fill slope is inclined at β = 24° (about 2.25H:1V) with φ′ = 36°, γ = 120 pcf. A shallow failure surface parallel to the slope at 6 ft depth is being checked. The seismic coefficient is kh = 0.12 applied horizontally (out of slope) at the block's center of mass. No pore pressure.

Most nearly, the pseudostatic factor of safety is:

A. 1.63
B. 1.46
C. 1.35
D. 1.22

Given equations / data

> Static infinite slope (ru = 0): F = tan φ′/tan β. Resolve W and khW parallel/normal to the slip plane: F = (cos β − kh sin β) tan φ′/(sin β + kh cos β)

Worked solution

1. Static F = tan36/tan24 = 1.63
2. Normal (per W): cos24 − 0.12 sin24 = 0.8647
3. Driving (per W): sin24 + 0.12 cos24 = 0.5164
4. F = 0.8647×tan36/0.5164 = 1.22 (depth and γ cancel)

Final answer

1.22

Why the other values are wrong

- **1.63** — static FS only
- **1.46** — divided static FS by (1 + kh)
- **1.35** — added kh·sin β to the normal force (wrong sign)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.4 Infinite Slope, p. 105; pseudostatic resolution as in §3.6.3 (kW terms); GEC 3 Ch. 7

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=SXA_TC3XEwk (no creator values copied).

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